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Evaluating calibration of deep fault diagnostic models under distribution shift

Abstract: Current intelligent fault diagnosis studies focus on improving model accuracy. While accuracy is crucial, an exclusive emphasis on this metric can leave users oblivious to potentially untrustworthy decisions made by the model. This underscores the importance of confidence estimation and brings the model miscalibration problem to the forefront, i.e., the softmax probability, which is supposed to indicate the likelihood of the predicted label being correct, fails to reflect the true probability accurately. Addressing this issue is imperative for several reasons. Firstly, a well-calibrated model can provide users with an assessment of the risk associated with prediction failures, thereby withholding decision-making when the confidence is low and mitigating the risk of erroneous outputs. Especially in situations involving out-of-distribution (OOD) and distribution-shifted inputs, where the risk of model failure increases, the calibration property becomes even more critical. Secondly, well-calibrated confidence estimates can enhance users' trust in today's many black-box models. However, there have been limited fault diagnosis studies that specifically explore model calibration. The effectiveness of existing calibration methods in handling out-of-distribution (OOD) and distribution-shifted inputs also remains unclear. Therefore, this paper evaluates multiple calibration methods and discusses their advantages and limitations, providing insights for subsequent studies. The results suggest that deep ensemble, as a method of deriving predictive expectations using multiple models with significantly different structures or parameters, has the potential to be the best calibration method. Code is available at **https://github.com/xiaoyiming1999/Calibration_for_RMFD**.

Key Words: Confidence estimation, distribution shift, fault diagnosis, model calibration, out-of-distribution sample, trustworthy artificial intelligence.

I Introduction

Since deep learning has been applied to mechanical fault diagnosis, many impressive results have been achieved in various diagnostic tasks such as few-shot fault diagnosis, cross-domain fault diagnosis, and fault diagnosis under strong noise interferences [1]-[4]. Recent research in intelligent fault diagnosis has predominantly focused on two strategies: designing robust network structures to extract detailed fault features [5]-[7] and developing comprehensive loss functions for model training **Error! Reference source not found.**, [9]. The overarching aim of these efforts is to enhance the diagnostic accuracy of models within specified scenarios. Notably, even approaches prioritizing interpretability in fault diagnosis claim that making diagnostic models interpretable inherently boosts their accuracy [10], [11].

Accuracy is undoubtedly crucial in mechanical fault diagnosis, a field primarily concerned with classification tasks. However, an exclusive focus on accuracy suggests an over-reliance on intelligent models in determining mechanical equipment failures. This reliance is concerning because users have no way of discerning the trustworthiness of the model's diagnostic decisions [12], [13]. From both ethical and safety standpoints, this approach is inadequate. A reliable fault diagnosis model should not only

provide accurate results but also indicate the likelihood of inaccuracies in its decisions. This brings the importance of confidence estimation to the forefront: a model should offer users a clear indication of the confidence level associated with its diagnostic results. High-confidence results can be a solid basis for decision-making, while results with lower confidence should prompt users to conduct a more comprehensive assessment of the equipment's state. At first glance, it might seem feasible to use the softmax output of a deep model directly as a measure of result confidence. However, recent studies [14]-[16] challenge this assumption, revealing that these highly accurate models, trained with negative log likelihood (NLL), often exhibit poor calibration. In other words, the probability indicated by the model's softmax output does not accurately represent the actual likelihood of the predicted label. A key manifestation of this problem is the model's tendency to make overconfident predictions, meaning it often overestimates the real-world correctness of its predicted labels. Inaccurate yet overconfident models can be misleading, highlighting the need for better-calibrated, trustworthy models in mechanical fault diagnosis.

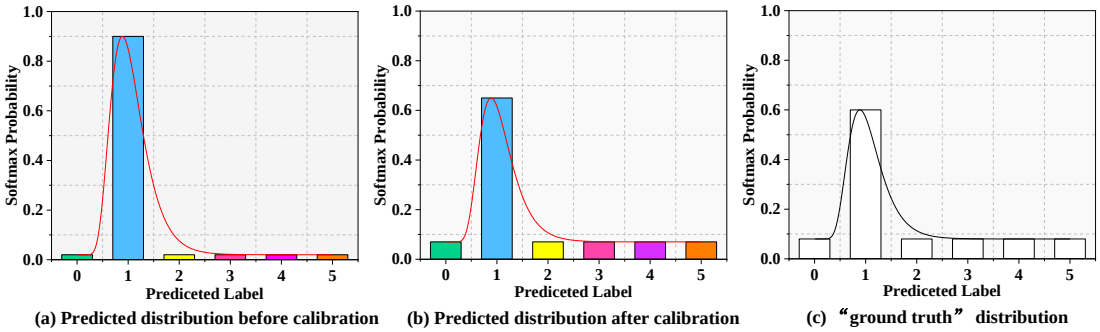


Fig. 1. The predicted distributions for a given sample before and after model calibration and the "ground truth" distribution of confidence estimate for that sample. Note that this "ground truth" distribution is usually unobservable and is approximated by empirical probability.

To align the predicted probability as closely as possible with the true likelihood, model calibration becomes a necessity. Note that calibration and accuracy are two orthogonal issues; a model with high accuracy may also have high calibration error, and vice versa. Research indicates that the susceptibility of deep models to overfitting, particularly due to their high capacity and the limited data available for training with the commonly used negative log likelihood (NLL) loss, is a primary cause of miscalibration [17]. This challenge has spurred the development of various calibration methods for image classification, such as temperature scaling [14], focal loss [17], and confidence penalty [18]. As shown in **Fig. 1**, these methods either explicitly or implicitly flatten the predicted distribution (softmax output) to prevent the model from being overconfident, while the difference between the predicted distribution and the "ground truth" distribution of the confidence estimate is reduced. Therefore, their application to calibrate diagnostic models is expected to facilitate more trustworthy fault diagnosis.

Furthermore, in the context of fault diagnosis, complex failure mechanisms may cause out-of-distribution (OOD) inputs with unknown fault class [19], **Error! Reference source not found.**, while variable noise backgrounds and operating conditions can lead to shifts in the distribution of the inputs [21], **Error! Reference source not found.**. A well-calibrated model in such scenarios would enable

users to accurately assess prediction failure risks and understand how accuracy degrades under these conditions. This enhances the robustness of the diagnostic model against OOD inputs as well as distribution-shifted inputs. In addition, considering the often-opaque decision-making logic of black-box deep models, calibration can enhance the user's trust in them. This is because humans have an innate cognitive intuition about probability, and reliable confidence estimates can provide a valuable reason for users to trust the diagnostic models [14]. However, to our knowledge, research on model calibration in mechanical fault diagnosis is still limited. Moreover, the above calibration methods, especially temperature scaling, are initially proposed mainly for scenarios where the training and test sets have similar distributions. Their effectiveness in handling OOD and distribution-shifted inputs, which are prevalent in fault diagnosis tasks, remains unclear. Therefore, this paper attempts to calibrate deep diagnostic models across various fault diagnosis tasks using these existing methods and discusses their advantages and limitations. The main contributions of our work are as follows:

(1) This research emphasizes the calibration of deep models in fault diagnosis. It stands apart from the typical focus on diagnostic accuracy prevalent in most current research, which not only highlights the importance of reliability and trust in diagnostic decision-making, but also adds a new dimension to existing research on mechanical fault diagnosis.

(2) The performance comparison of various calibration methods is not limited to the in-distribution (ID) test samples commonly used in previous model calibration studies, but focuses on OOD samples and distribution-shifted samples that are commonly encountered in mechanical fault diagnosis. The results suggest that deep ensemble has the potential to be the best calibration method.

(3) We analyze in detail the behavior of various types of calibration methods and derive several valuable conclusions. Meanwhile, we release an algorithm that deploys these evaluated methods. They can provide insights into subsequent calibration studies and lay a solid foundation for their advancement. Our code is available at **https://github.com/xiaoyiming1999/Calibration_for_RMFD**.

The remainder of this article is organized as follows: Section II defines the model calibration problem. Section III introduces the calibration methods used for the comparative studies and their hyperparameter selection. Section IV describes the experimental setup and analyzes the results. The conclusions and prospects are discussed in Section V.

II Problem Statement

Let x be an input and $y \in \{0,1,...,K-1\}$ be its ground truth label, and $f(\cdot)$ be a deep neural network with $f(x) = (\hat{y}, \hat{p})$, where K is the number of known classes, $\hat{y}$ is the predicted label of x, and $\hat{p}$ is the softmax probability associated with $\hat{y}$. We would like the estimated confidence $\hat{p}$ to be representative of the true probability that the predicted label $\hat{y}$ is correct. For example, given 100 predictions with a confidence of 0.8, 80 of them should be correct. Therefore, a perfectly calibrated model should satisfy

$$P(\hat{y} = y \mid \hat{p} = p) = p, \forall p \in [0,1] \tag{1}$$

Considering that $\hat{p}$ is a continuous random variable and the number of inputs is usually finite, it is difficult to evaluate the calibration quality of the model directly using **Eq. (1)**. This motivates a number

of approximate metrics that capture the essence of **Eq. (1)**.

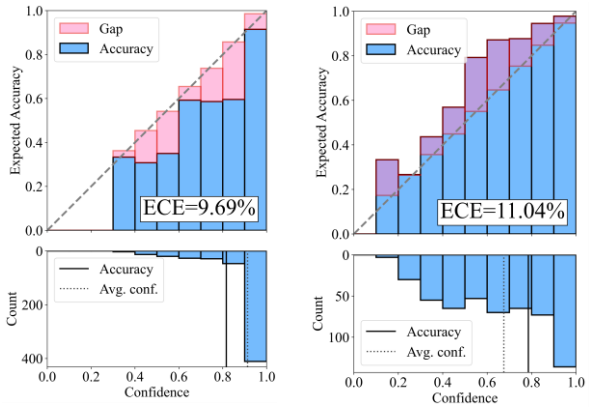
Reliability diagram is an important visualization tool for calibration quality that describes accuracy as a function of confidence, as shown in **Fig. 2** top. Specifically, we divide the confidence interval [0,1] into *M* equally sized intervals, each associated with a bin, and place the predictions into the corresponding bins according to their confidence. Let B_m be the bin associated with the confidence interval $I_m = (\frac{m-1}{M}, \frac{m}{M}]$. The accuracy of the predictions within B_m is

$$\mathrm{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}(\hat{y}_i = y_i) \tag{2}$$

where i is the data index, and $\mathrm{acc}(B_m)$ is an unbiased estimate of $P(\hat{y} = y \mid \hat{p} \in I_m)$. In addition, the averaged confidence of the predictions within B_m is

$$\mathrm{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i \tag{3}$$

$\mathrm{acc}(B_m)$ and $\mathrm{conf}(B_m)$ can be regarded as approximations of the left and right sides of **Eq. (1)**, respectively. Therefore, a perfectly calibrated model, i.e., satisfying **Eq. (1)**, should have $\mathrm{acc}(B_m) = \mathrm{conf}(B_m)$ for any $m \in \{1, 2, ..., M\}$, which makes its reliability diagram present as an identity function (diagonal lines in **Fig. 2** top). As shown in **Fig. 2** top, any deviation of the blue bar from the diagonal represents the miscalibration, where a deviation below the diagonal is model overconfidence and a deviation above the diagonal is model underconfidence. Note that a reliability diagram alone cannot show the number of samples in each bin, and thence it cannot be used to evaluate the number of samples being calibrated, which requires a confidence histogram, as shown in **Fig. 2** bottom.



(a) Overconfident predictions **(b) Underconfident predictions**

Fig. 2. Reliability diagrams (top) and confidence histograms (bottom) with 10 bins, where the blue bars on the top and bottom represent the accuracy and number of predictions within each bin, respectively, the red bars denote the calibration gap, the diagonal lines represent the perfect calibration, and the vertical solid and vertical dashed lines are the accuracy and the averaged confidence of the model for all samples, respectively.

Expected calibration error (ECE) clearly evaluates the calibration quality of the model with a scalar compared to reliability diagram as a visualization approach. Mathematically, ECE is defined as the expected difference between accuracy and confidence

$$\mathop{\mathbb{E}}_{\hat{p}}\left[\left|P(\hat{y}=y \mid \hat{p}=p)-p\right|\right] \tag{4}$$

In practice, similar to reliability diagram, we assign the predictions to M equally sized confidence bins and take a weighted average of the difference between the accuracy and confidence of all the bins

$$\mathrm{ECE}=\sum_{m=1}^{M}\frac{\left|B_m\right|}{n}\left|\mathrm{acc}(B_m)-\mathrm{conf}(B_m)\right| \tag{5}$$

where n is the number of samples. The difference between acc and conf (red bar in **Fig. 2** top) for each bin indicates the calibration gap. Note that although ECE can quantitatively show the degree of model miscalibration, it cannot determine whether the model is overconfident or underconfident.

Moreover, there are some metrics such as NLL and Brier score [15] that are commonly used to evaluate probabilistic models. In the context of deep learning, NLL is also known as cross-entropy loss, and Brier score is the squared error between the softmax probability vector and the one-hot label. Note that the confidence of prediction is not an inherent property of itself like the label of the sample. Different models may not have the same confidence in their predictions for the same sample, but they may all be well-calibrated.

III Calibration Methods Evaluated

This paper categorizes existing calibration methods into four groups: post-hoc methods, regularization methods, data augmentation methods, and uncertainty estimation methods. Each of these approaches offers distinct mechanisms to enhance model calibration. Specifically, post-hoc methods recalibrate the predicted probabilities by scaling the logits of the model on a held-out validation set, with the most state-of-the-art post-hoc method known as temperature scaling. Regularization methods [17], [18] regularize the NLL loss to increase the entropy of the predicted distribution while ensuring that the Kullback-Leibler (KL) divergence between it and the target distribution (one-hot encoding) is minimized. Data augmentation methods [23]-[25] reduce the confidence of a model in its predictions by adding noise to the training samples or their ground truth labels. Uncertainty estimation methods are mainly based on Bayesian variational inference [26]-[28] or deep ensemble [29], [30], which aim to marginalize over the model parameters to alleviate the impact of epistemic uncertainty on the results and thus produce well-calibrated predictors.

This paper selects 7 calibration methods (covering all the above-mentioned types of calibration methods) and a benchmark method for the comparative study according to their prevalence, scalability, and applicability. The network architecture used in all methods except deep ensemble is Reset18. The main hyperparameters are set as follows: the batch size is 64, the number of epochs is 30, the weight decay is 1×10^{-5}, and the initial learning rate is 1×10^{-3} with a decay (multiplied by 0.1) in epoch 15 and epoch 25. These methods include:

Vanilla, a benchmark method to train deep model using only standard cross-entropy loss.

Temperature scaling (TS) is a post-hoc method that uses only one temperature parameter learned on a held-out validation set to scale the logits of deep model. TS can usually be used in combination with other

types of calibration methods to further enhance their performance.

Focal loss (FL), a regularization method, was first proposed for addressing class imbalance issue and subsequently shown to be equivalent to a trade-off between minimizing KL divergence and maximizing entropy. The performance of FL depends on its hyperparameter α, which is selected as 3 in this paper based on the experimental results in Section IV-A.

Confidence penalty (CP) is a regularization method that adds an entropy maximization term directly to the standard cross-entropy loss and uses a hyperparameter β to control the strength of the confidence penalty. In this paper, β is selected as 0.15 based on the experimental results in Section IV-A.

Label smooth (LS) [23] is a data augmentation method that introduces a smoothing parameter γ to the standard cross-entropy loss to soften one-hot labels and uses the softened labels to calculate loss. In this paper, γ is set to 0.01 based on the experimental results in Section IV-A.

Mixup [24] is a data augmentation method that combines two samples by interpolating both the inputs and labels to generate an augmented input and its label, where the combination ratio of the two samples is drawn from a Beta distribution. Typically, the two shape parameters of the Beta distribution are set to 1, and this setting is followed in this paper.

Monte Carlo Dropout (MC Dropout) [26] is an uncertainty estimation method that not only uses dropout layers in training phase, but also activates them in inference phase and performs multiple forward propagations to find the expectation of prediction. Since the use of dropout layer is equivalent to treating the network parameters as random variables obeying a Bernoulli distribution, MC dropout can be approximated as performing variational inference on the posterior distribution of the network parameters. However, the original dropout layer is only applicable to fully connected layers, whereas fault diagnostic models are usually based on convolutional neural networks (CNN), where the fully connected layers make up only a relatively small part of the network. This causes that the dropout may not be sufficient to introduce enough randomness into the network parameters, which in turn results in poor calibration [28]. Therefore, we instead use a dropout variant designed for convolutional layers, dropblock [31], whose hyperparameters *keep_prob* and *block_size* are set to 0.9 and 8, respectively, with reference to the experimental results in [31]. The number of Monte Carlo sampling in inference phase is set to 16. This parameter should not be too small or it will lead to a biased expectation of prediction; nor should it be too large or it will lead to a high computational cost.

Deep ensemble (DE) is an uncertainty estimation method that uses multiple models with significantly different structures or parameters to derive the expectation of prediction. In some previous studies [15], [30], [32], DE has even been considered the most promising calibration method. In this paper, we use 5 models as members of the ensemble, including a Resnet18, a 5-layer CNN, a 7-layer CNN, a bi-directional long short-term memory, and a multiscale dilated convolutional network [6]. Moreover, similar to [29], each model is trained not only using the standard cross-entropy but also the fast gradient sign method [33] for adversarial training, where the perturbation ε applied to the original inputs follows the setting in [29] and is set to 0.01. This not only further smoothes the predicted distribution, but also further enhances the model diversity because each model is trained using different adversarial samples, which contributes to the calibration quality [28].

IV Experimental Study

Previous model calibration studies usually assume that the training and test sets are sampled from the same joint distribution $P(X,Y)$, where X is the union of inputs, and Y is the union of labels. This paper considers not only this scenario but also two common scenarios in the context of fault diagnosis where the test set follows a different distribution $P'(X,Y)$: (1) $P(X,Y) \neq P'(X,Y)$ due to $P(Y) \neq P'(Y)$. Complex failure mechanisms can result in OOD samples whose ground truth label is not one of the known K classes. For this scenario the evaluation measures in Section II will no longer apply. We would like that the calibrated model's predicted distributions for the OOD samples is flatter than its predicted distributions for the ID samples. Furthermore, it should be possible to effectively distinguish between ID and OOD samples based on the magnitude of entropy of the predicted distribution. (2) $P(X,Y) \neq P'(X,Y)$ due to $P(X) \neq P'(X)$. Varying noise backgrounds can cause training data and test data to have different signal-to-noise ratios (SNR). In addition, varying equipment speeds and loads can make the training and test sets statistically different. Such distribution-shifted samples will degrade the model accuracy, and we would like the model predictions to become less confident as the shift is enhanced, thus keeping a low calibration error. This section includes three experimental cases, where the test samples in Case 1 include ID and OOD samples, the test samples in Case 2 are increasingly shifted samples as the noise is enhanced, and the test samples in Case 3 are increasingly shifted samples as the equipment rotational speed decreases.

IV-A Case 1: Bearing Fault Diagnosis for ID and OOD Samples

***1) Data preparation*:** The dataset used in Case 1 is a bearing fault dataset provided by Paderborn University (PU) [34]. **Fig. 3** shows their experimental platform with a piezoelectric accelerometer mounted on the bearing housing to collect vibration signals at a sampling frequency of 64KHz. There are 32 test bearings, including 6 normal bearings, 14 real damaged bearings, and 12 artificially damaged bearings. In addition, there are 4 different operating conditions, each of which consists of a specific motor speed, radial force on the bearing, and load on the drive system.

This paper uses the condition monitoring signals of 14 real damaged bearings collected under operating condition of speed of 1500rpm, radial force of 400N, and load of 0.7N·M to conduct the comparative studies. The details of these 14 bearings are listed in **Table I**. We use the data labeled 0, 1, 2, ..., 10 to construct training and test samples, while data labeled 11, 12, and 13 are used only to construct test samples to serve as OOD samples with unknown fault categories. There are a total of 4400 (400×11) training samples, and 700 (50×14) test samples, each with a length of 1024. To improve the challenge of the diagnostic tasks, the SNR is made 5dB by applying Gaussian noise to each sample. The zero-mean normalization is performed on each sample in all methods except in DE where the min-max normalization is performed on each sample to facilitate the setting of the perturbation ε.

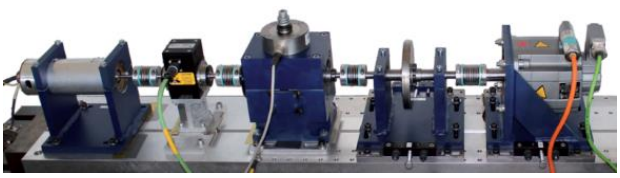


Fig. 3. PU experimental platform.

Table I

Details of 14 Real Damaged Bearings

Bearing Code	Damage mode	Damaged element	Combination	Damage Characteristic	Label
KA04	M1	OR	S	C1	0
KA15	M2	OR	S	C1	1
KA16	M1	OR	R	C1	2
KA22	M1	OR	S	C1	3
KA30	M2	OR	R	C2	4
KB23	M1	IR (+OR)	M	C1	5
KB24	M1	IR (+OR)	M	C2	6
KB27	M2	OR+IR	M	C2	7
KI04	M1	IR	M	C1	8
KI14	M1	IR	M	C1	9
KI16	M1	IR	S	C1	10
KI17	M1	IR	R	C1	11
KI18	M1	IR	S	C1	12
KI21	M1	IR	S	C1	13

M1: fatigue, pitting; M2: plastic deform, indentations; OR: outer ring; IR: inner ring; S: single damage; R: repetitive damage; M: multiple damages; C1: single point; C2: distributed.

Table II

Results for ID samples in Case1

Methods	Accuracy (%)	ECE (%)	NLL	Brier score	Time (ms)
Vanilla	81.23	9.59	0.69	0.29	160
TS	81.20	4.12	0.55	0.27	160
FL	83.24	3.34	0.47	0.24	160
CP	81.13	4.00	0.55	0.26	160
LS	77.27	3.36	0.69	0.32	160
Mixup	79.30	11.24	0.70	0.32	160
MC dropout	86.14	3.02	0.41	0.21	3210
DE	86.91	2.76	0.36	0.19	220

***2) Result analysis*: Table II** lists the experimental results of each method for the ID samples, where each result is the average of the results of five replicate trials. **Fig. 4** shows the reliability diagrams and confidence histograms at a certain trial. As listed in **Table II**, Vanilla has the highest ECE except for Mixup. Furthermore, the confidence of samples in Vanilla is overwhelmingly in the (0.9, 1) interval

(higher than the accuracy), as shown in **Fig. 4**, further reflecting the fact that deep models are prone to be overconfident. The accuracy of TS is remarkably close to the accuracy of Vanilla, and its ECE is lower than Vanilla's ECE. This is because TS simply scales the logits of Vanilla linearly to flatten its predicted distributions, and thus does not change its classification results while reducing its confidence.

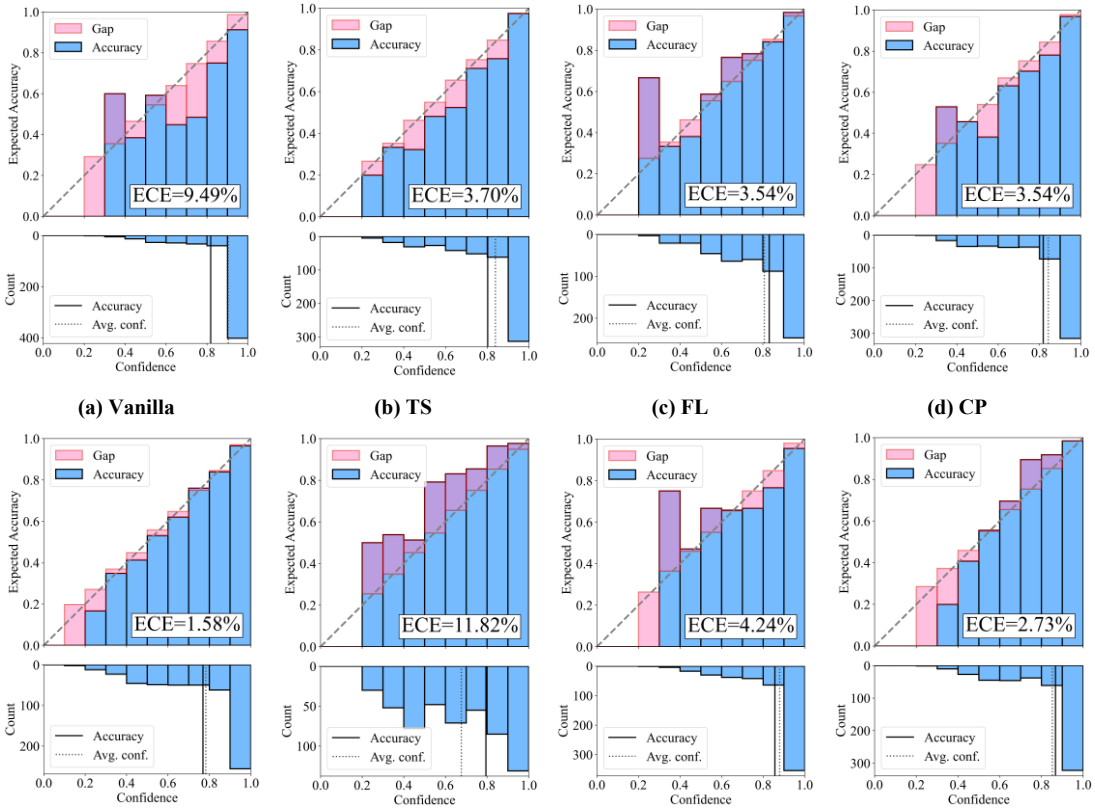


Fig. 4. Reliability diagrams and confidence histograms (10 bins) for ID samples in Case 1.

Both FL and CP have decreased ECE compared to Vanilla's ECE. Moreover, there is an increase in the accuracy of FL compared to Vanilla's accuracy. These results indicate that they have good calibration capability. However, it should be mentioned that the performance of FL and CP is highly dependent on their respective hyperparameters α and β, as shown in **Fig. 5a** and **5b**. If the values of α and β are chosen too high, the predicted distributions of the model may become too flat, resulting in an excessive lack of confidence in the model for correct results. Conversely, if the values of α and β are chosen too low, it may even be difficult to weaken the model's confidence for incorrect results. To consider both accuracy and calibration quality, this paper sets α and β to 3 and 0.15, and uses them as examples to analyze their effects. As can be seen in **Figs. 4a**, **4c**, and **4d**, the choice of α=3 makes the number of low-confidence samples (samples with confidence below accuracy) in FL significantly higher than that in Vanilla. These samples may include some samples that are correctly predicted but whose confidence has been excessively lowered, which causes the model to change from overconfident to underconfident. In addition, the choice of β=0.15 makes the increase in the number of low-confidence samples in CP less compared to the increase in FL, and the model remains overconfident. Interestingly, both FL and CP have an ECE of 3.54%, but they are in two opposite states.

Although LS reduces the ECE of the deep model, it also comes at the cost of sacrificing model accuracy (e.g., 77.27% in **Table II**). This is because models for fault diagnosis are usually of low capacity (e.g., Resnet18 in this paper) and the number of fault classes to be diagnosed is small. In this case, LS usually struggles to perform its well-known role of improving accuracy, and is even harmful to model accuracy [23]. In addition, due to the significant decrease in accuracy, the Brier score of LS is even worse than that of Vanilla. Therefore, it can be assumed that the calibration quality of LS in Case 1 is lower than that of Vanilla. The performance of LS also depends on its hyperparameter γ. As can be seen in **Fig. 5c**, as γ increases, the ECE of LS decreases and then increases, which indicates that LS gradually reduce its confidence in results, causing its state to change from overconfidence to underconfidence. Furthermore, as shown in **Fig. 4e**, although LS misclassifies more samples, it correspondingly reduces their confidence, so that the number of low-confidence samples in LS is significantly higher than that in Vanilla, and it achieves an ECE of only 1.58%. The ECE of Mixup in this case is even higher than the ECE of Vanilla. As can be seen in **Fig. 4f**, Mixup can produce a large number of low-confidence samples, and among these are a large number of correctly predicted samples, leading to a severe lack of confidence in the model.

DE achieves the highest accuracy, the lowest ECE, the lowest NLL, and the lowest Brier score. In addition, the accuracy of MC dropout is only slightly lower than that of DE, and its ECE, NLL, and Brier score are all only slightly higher than that of DE. These results demonstrate that the uncertainty estimation methods have excellent classification and calibration performance when handling ID samples. However, it is important to note that they tend to be more computationally expensive. **Table II** lists the time required by each method to inference over all ID test samples. It can be observed that all methods except the uncertainty estimation methods take the same amount of time because they do not introduce additional computational cost. However, the computational cost of MC dropout increases dramatically due to the requirement for multiple forward propagations in the inference phase. Although the computational cost of DE is improved by 37.5% compared to that of the non-uncertainty estimation methods, this cost increase is acceptable considering the performance enhancement it brings. It is worth mentioning that the DE in this paper uses adversarial attack rather than relying solely on differences in model structure to enhance model diversity, which may be the key to the fact that it does not require introducing excessive model parameters but still has excellent performance.

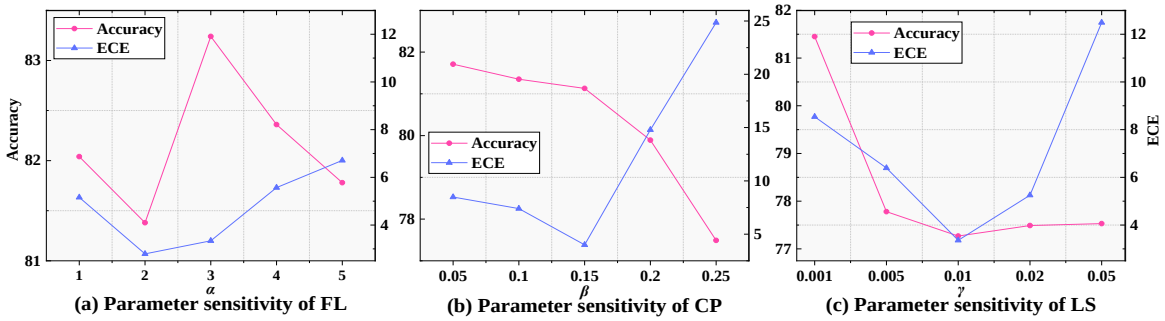


Fig. 5. Parameter sensitivity analysis of methods.

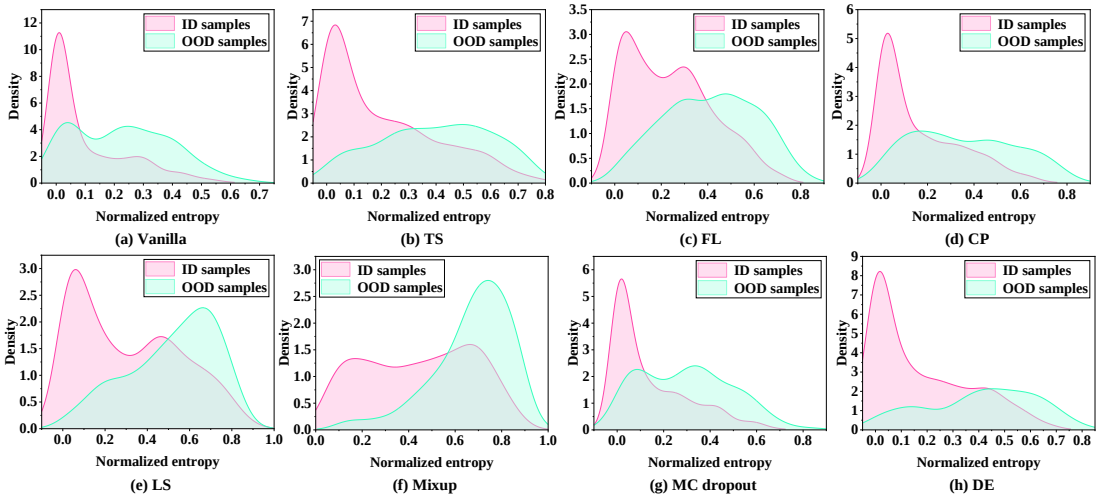


Fig. 6. Density of normalized entropy of predicted distributions for ID and OOD samples in Case 1.

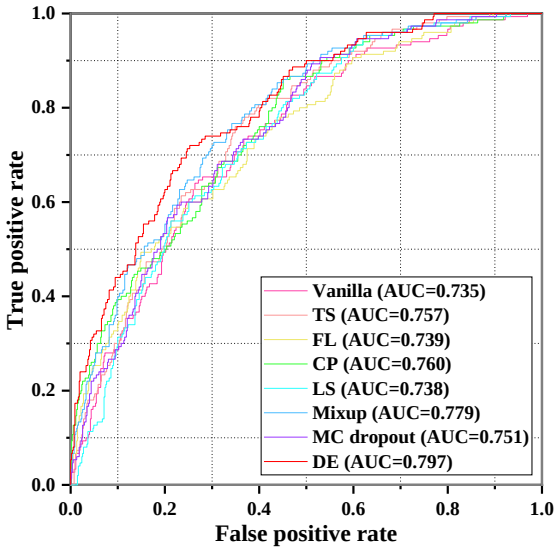


Fig. 7. ROC curves for ID and OOD samples in Case 1.

Fig. 6 presents the density of the normalized entropy of predicted distributions of each method for ID and OOD samples, where the normalized entropy of the predicted distribution is defined as the ratio of its entropy to maximum value $\log K$. **Fig. 7** shows the receiver operating characteristic (ROC) curve and the value of area under ROC curve (AUC) for each method to discriminate between ID and OOD samples based on the magnitude of the normalized entropy. Intuitively, calibrated deep models should lack confidence in handling OOD samples, and therefore the entropy of OOD samples should be significantly higher than the entropy of ID samples. However, in Case 1, there are also a number of incorrectly predicted ID samples with low confidence that also have high entropy. In addition, the OOD samples may have similar features to the samples with a certain known label, causing the model to assign a relatively high predicted probability to that known label when processing these samples. Thus, there is an overlap between the density plot of the ID samples and that of the OOD samples for each method in **Fig. 6**.

Since the main intervals in which the entropy of ID samples and that of OOD samples in other methods except Mixup do not seem to show significant differences, we try to analyze the reasons for the performance differences in OOD detection by methods from the perspective of the number of low-confidence ID samples in each method. As shown in **Fig. 6**, for Vanilla, the density plot of the ID samples is more peaked, and their normalized entropy values are mainly concentrated in the interval (0, 0.1). In contrast, the density plots of TS, CP, MC dropout, and DE for the ID samples are relatively flat, and those of FL, LS, and Mixup are quiet flat, all of which are consistent with the previous analysis. Since there are more low-confidence ID samples in FL and LS, it is difficult for them to distinguish between ID samples and OOD samples, and their AUC values are quite close to the AUC of Vanilla, as shown in **Fig. 7**. However, although there are also a large number of low-confidence ID samples in Mixup, the entropy it assigns to the OOD samples is significantly higher, resulting in a 4.4% increase in its AUC compared to Vanilla's AUC. For TS, CP, and MC dropout, they neither excessively reduce the confidence of correctly predicted samples as FL does, nor do they produce more incorrectly predicted samples as LS does, and thus their increase in the number of low-confidence ID samples is smaller, and their AUC are all improved compared to Vanilla's AUC. For DE, it has the highest accuracy and calibration quality, and therefore the confidence it assigns to ID samples is generally higher, resulting in the highest AUC.

IV-B. Case 2: Gearbox Fault Diagnosis for Increasingly Shifted Samples Due to Noise Enhancement

***1) Data preparation*:** The dataset used in Case 2 is a planetary gearbox dataset provided by Southeast University (SEU) [35]. **Fig. 8a** shows their experimental platform with a three-channel accelerometer to collect vibration signals at a sampling frequency of 12KHz under an operating condition of 2500rpm motor speed. There are 13 health states of gearbox, including normal and 12 failure states as shown in **Fig. 8b**. The details of these 13 health states are listed in **Table III**.

This paper selects the signals from the first channel to construct the samples. There are a total of 2600 (200×13) training samples and 650 (50×13) test samples, each with a length of 1024. By applying Gaussian noise to these samples, the SNR of each training sample is fixed at 5dB, and the SNR of each test sample is gradually reduced from 5dB to 0dB at 1dB intervals, thus serving as the increasingly shifted samples due to noise enhancement. The pre-processing methods used on the samples are consistent with those used in Case 1.

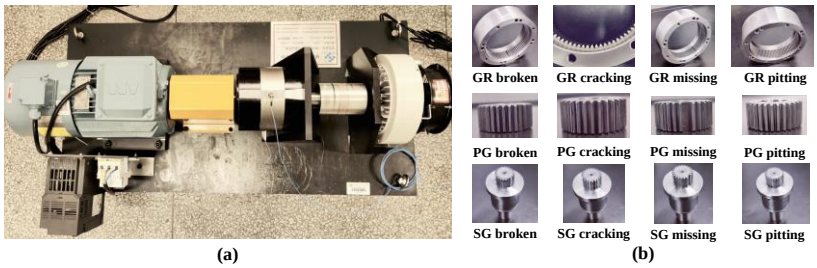


Fig. 8. SEU experimental platform.

Table III

Description of 10 Health States of Gearbox

States of gearbox	Failure element	Description	Label
Normal	-	-	0
GR broken	Gear ring	Tooth broken	1
GR cracking	Gear ring	Tooth crack	2
GR missing	Gear ring	Tooth missing	3
GR pitting	Gear ring	Pitting	4
PG broken	Planetary gear	Tooth broken	5
PG cracking	Planetary gear	Tooth crack	6
PG missing	Planetary gear	Tooth missing	7
PG pitting	Planetary gear	Pitting	8
SG broken	Sun gear	Tooth broken	9
SG cracking	Sun gear	Tooth crack	10
SG missing	Sun gear	Tooth missing	11
SG pitting	Sun gear	Pitting	12

***2) Result analysis*: Fig. 9** illustrates the experimental results of each method for the increasingly shifted samples due to noise enhancement, where each result is the average of the results of five replicate trials. **Fig. 10** shows the reliability diagrams and confidence histograms for the test samples with a SNR of 2dB at a certain trial. As can be seen in **Fig. 9a**, as the shift of the test samples increases, the accuracy of each method inevitably decreases gradually, among which DE has the smallest decrease, demonstrating its strong robustness and noise resistance.

Compared to ECE plot (**Fig. 9b**), Brier score plot (**Fig. 9c**) shows more clearly the trend of the calibration quality of each method with the increasing shift and disambiguate the evaluation of their calibration performance. As shown in **Fig. 9c**, the Brier score of each method shows a clear upward trend, indicating that the calibration quality consistently degrades with increasing shift, regardless of method. The increase in Brier score for DE is the smallest of all the methods, again demonstrating its superiority. In addition, despite the poor performance of LS and Mixup when processing weakly shifted samples, they show better calibration quality than other methods except DE when the test samples are strongly shifted. Meanwhile, although the performance of MC dropout is second only to DE when the shift of the test samples is small, its calibration quality drops severely or even approaches the baseline Vanilla when the test samples are strongly shifted. These results suggest that better calibration for weakly shifted samples does not necessarily lead to better calibration for strongly shifted samples.

Further, we analyze the state changes with increasing shift for each method and the reasons for these changes via **Figs 9b** and **10**. As the shift increases, Vanilla, TS, CP, LS, MC dropout, and DE all have varying degrees of increased ECE and they become increasingly overconfident. This is because their accuracy decreases with increasing shift, but they do not correspondingly decrease the confidence of the predictions, both correct and incorrect. As a simple example, a perfectly calibrated predictor gives 100 predictions with a confidence of 80%, 80 of which should be correct, i.e., the accuracy is 80%. However, due to the increased shift, the accuracy of these 100 predictions becomes 70%. For this predictor to remain perfectly calibrated,

the confidence of these 100 predictions can to be reduced to 70%. Intuitively, samples that the predictor was confidently predicting now have to be less confident due to the samples being shifted. It is worth mentioning that we find that TS does not seem to lose its calibration ability for shifted samples as it does in the image classification tasks, and its ECE still shows a significant decrease compared to that of Vanilla.

Unlike the previous methods, the ECEs of both FL and Mixup experience a downward followed by an upward trend, i.e., their states change from underconfidence to overconfidence. This is because, as described in Case 1, when the samples are not shifted, FL (α=3) and Mixup are more accurate but tend to assign relatively low confidence to the predictions, resulting in models that are underconfident. As the shift increases, the accuracy of FL and Mixup gradually decreases, and so does the degree of model underconfidence. However, when the shift increases further and the accuracy of FL and Mixup decreases further, but they do not reduce the confidence for the predictions by as much as the decrease in accuracy, they become overconfident.

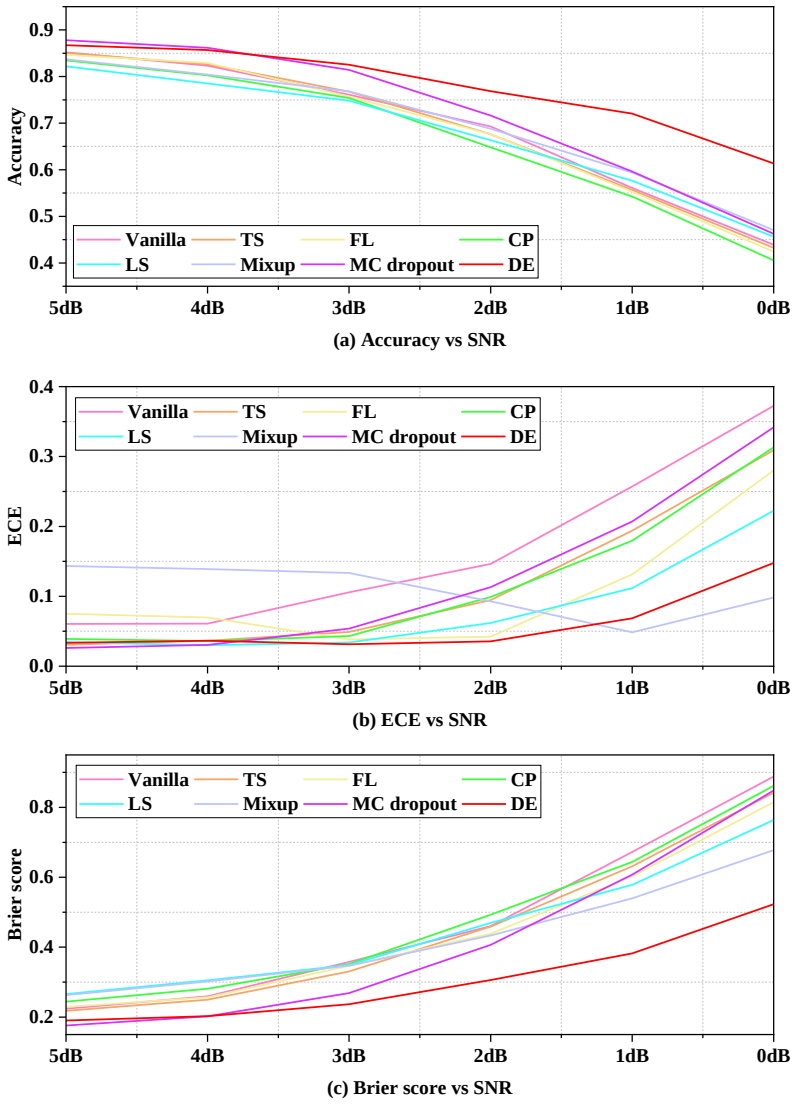


Fig. 9. Results for increasingly shifted samples in Case 2.

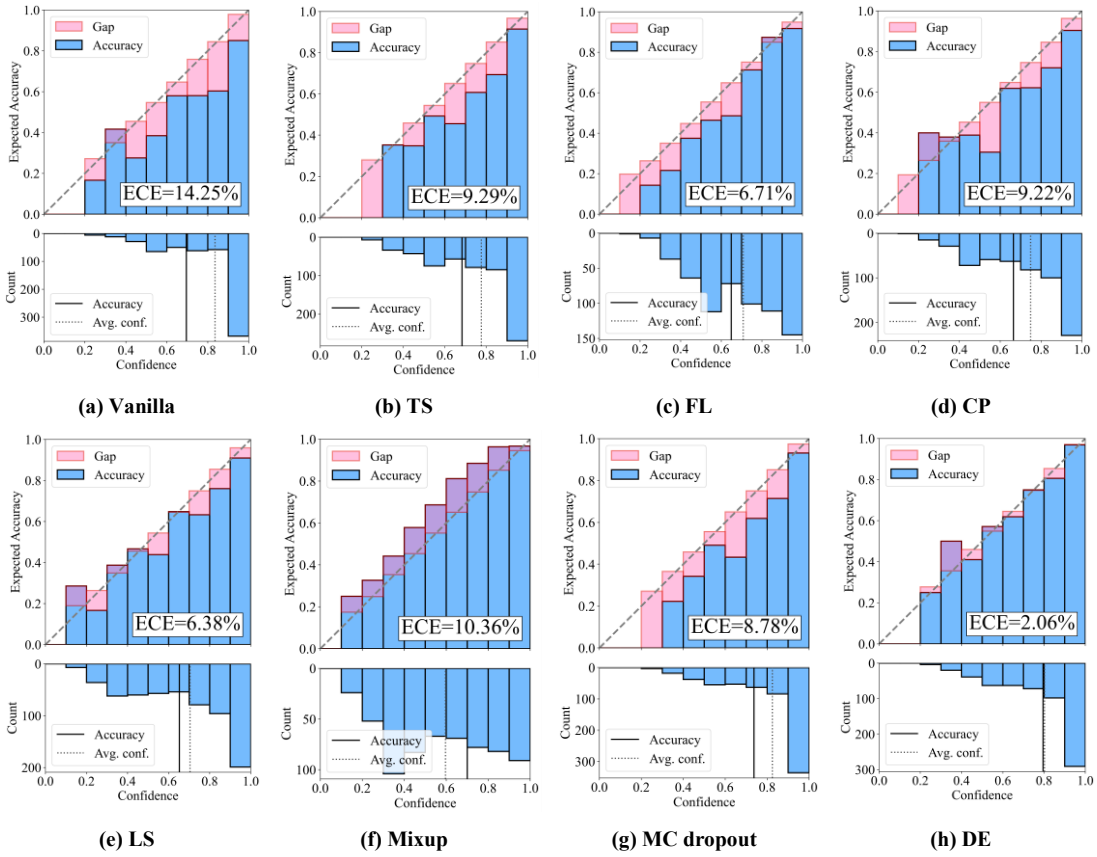


Fig. 10. Reliability diagrams and confidence histograms (10 bins) for test samples with a SNR of 2dB in Case 2.

IV-C. Case 3: Gearbox Fault Diagnosis for Increasingly Shifted Samples Due to Speed Reduction

***1) Data preparation*:** The dataset used in Case 3 is a planetary gearbox dataset for wind turbine provided by Tsinghua University (THU) [36]. **Fig. 11** shows their experimental platform, whose input is driven by a motor and the output is connected to a wind turbine. The gearbox casing is equipped with two acceleration sensors to collect vibration signals from its X and Y directions at a sampling frequency of 20KHz. There are 9 health states of gearbox, including normal, 4 sun gear failures, and 4 planetary gear failures, the details of which are listed in **Table IV**. The motor speed changes from 15Hz to 40Hz in 1Hz intervals, so there are 26 different operating conditions.

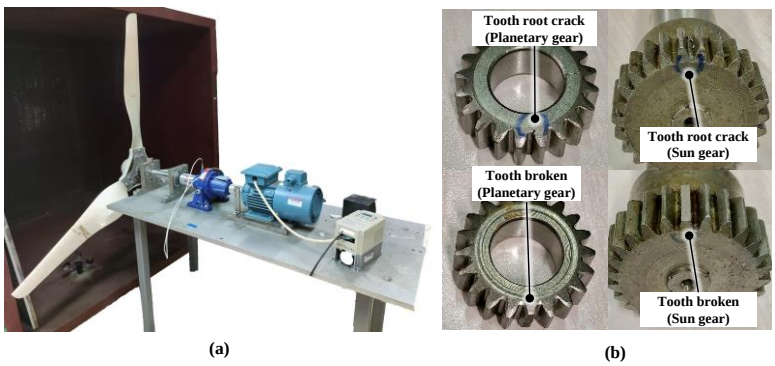


Fig. 11. THU experimental platform.

Table IV

Description of 9 Health States of Gearbox for Wind Turbine

Health states	Description	Label
Normal	-	0
Sun gear tooth crack	Depth is 1/8 of the tooth root width	1
Sun gear tooth crack	Depth is 1/4 of the tooth root width	2
Sun gear tooth crack	Depth is 1/2 of the tooth root width	3
Sun gear tooth broken	Cut 1/3 of the tooth height	4
Planetary gear tooth crack	Depth is 1/8 of the tooth root width	5
Planetary gear tooth crack	Depth is 1/4 of the tooth root width	6
Planetary gear tooth crack	Depth is 1/2 of the tooth root width	7
Planetary gear tooth broken	Cut 1/3 of the tooth height	8

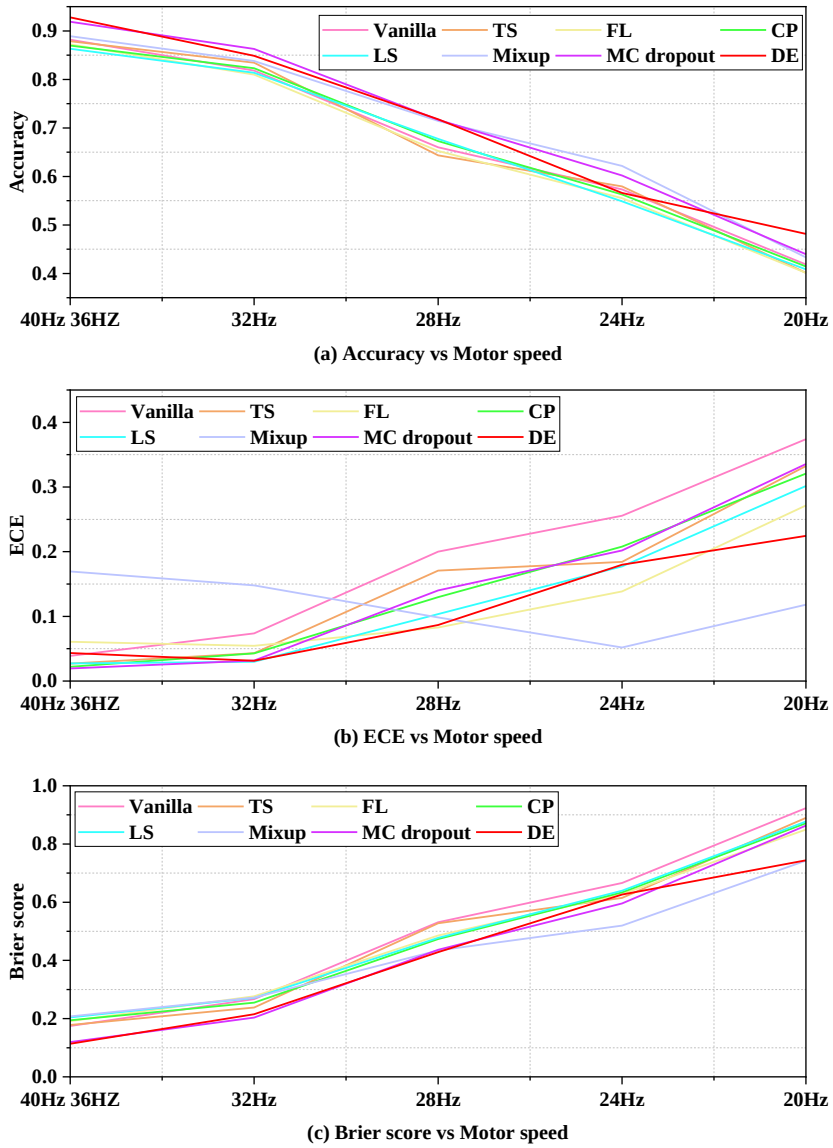


Fig. 12. Results for increasingly shifted samples in Case 3.

We use the signals from the X direction to conduct the comparative studies, where the signals with motor speeds of 40Hz and 36Hz are used to construct the training and test samples, and the signals with speeds of 32Hz, 28Hz, 24Hz, and 20Hz are used sequentially to construct the test samples to serve as the increasingly shifted samples due to speed reduction. There are a total of 900 (100×9) training samples (if available) and 450 (50×9) test samples for each operating condition, where the length of each sample is 1024. We apply Gaussian noise to each sample so that its SNR is 1dB to improve the task challenge. The pre-processing methods used on the samples are consistent with those used in Case 1.

***2) Result analysis*: Fig. 12** illustrates the experimental results of each method for the increasingly shifted samples due to speed reduction, where each result is the average of the results of five replicate trials. **Fig. 13** shows the reliability diagrams and confidence histograms for the test samples with a motor speed of 24Hz at a certain trial. Similar to Case 2, as the shift of the test sample increases, the accuracy of all methods shows a decreasing trend, while their BSs all show an increasing trend. Although the difference in performance presented by the methods in processing the increasing shifted samples in this case is not as obvious as in Case 2, it can still be inferred that the classification and calibration performance of DE is relatively better. In addition, MC dropout has high accuracy and calibration quality when processing weakly shifted samples. Finally, Mixup is an effective method for calibrating models that process strongly shifted samples.

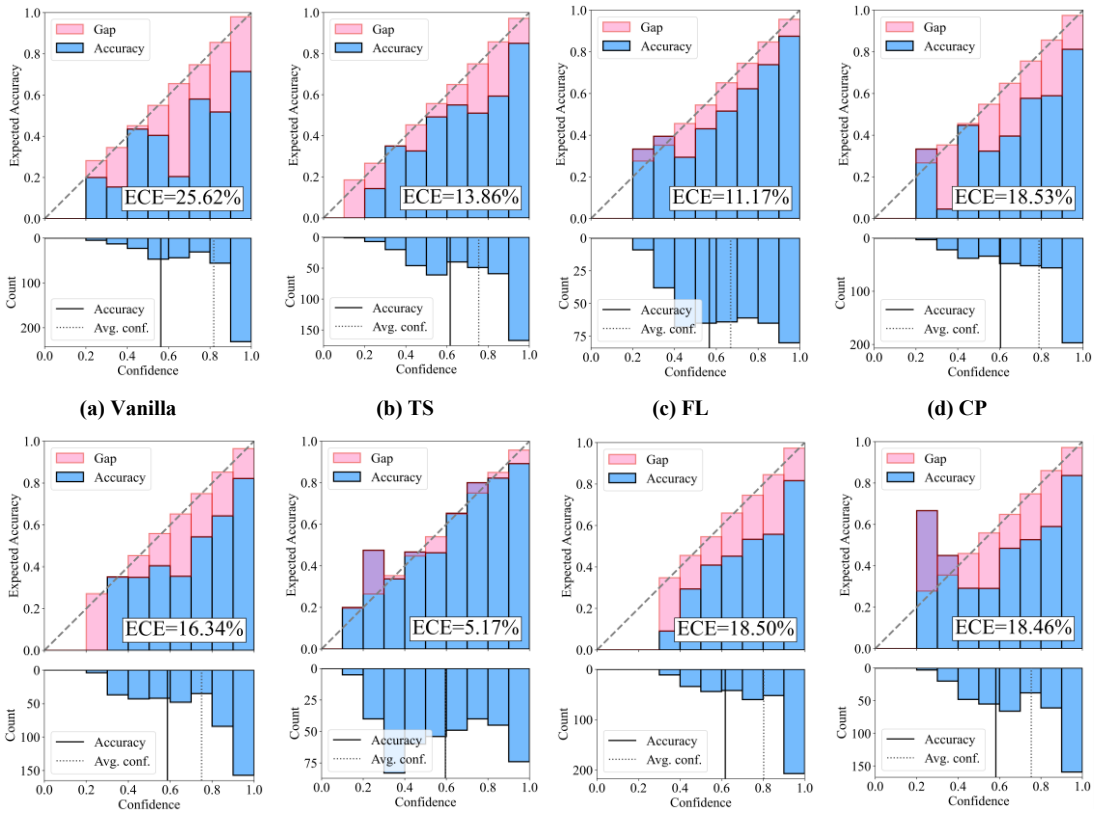


Fig. 13. Reliability diagrams and confidence histograms (10 bins) for test samples with a motor speed of 24Hz in Case 3.

V Conclusion

This paper begins by proposing that attention should be given to improving calibration in addition to accuracy, thus adding a new dimension to existing research on mechanical fault diagnosis. Subsequently, this paper suggests that it should not be limited to using ID test samples to evaluate the performance of calibration methods, but also OOD samples and distribution-shifted samples, which are common in mechanical fault diagnosis. Finally, this paper tests the performance of various calibration methods in three case studies and analyzes in detail their calibration behavior in each scenario. Based on our analysis, the following important conclusions can be deduced:

(1) The calibration quality consistently degrades with increasing shift, regardless of method and cause of shift.

(2) The better calibration for ID samples does not necessarily lead to better calibration for shifted samples, and vice versa.

(3) The number of low-confidence ID samples in a method is a key to its OOD detection performance.

(4) DE exhibits the best classification and calibration performance as well as the highest robustness to distributional shifts in most cases, suggesting that DE has the potential to be the best calibration method in fault diagnosis.

(5) In the context of fault diagnosis, TS does not lose its calibration ability for shifted samples as it does in the image classification tasks.

(6) LS can be used for model calibration in fault diagnosis, but usually at the expense of accuracy.

(7) MC dropout is a relatively good choice when dealing with weakly shifted samples.

(8) Mixup is applicable to calibrate models when processing strongly shifted samples.

(9) The calibration performance of FL and CP is highly dependent on their respective hyperparameters, making it difficult to achieve model calibration in various diagnostic tasks simultaneously.

Future work lies in three main areas: (1) Firstly, we plan to develop DE methods that are more versatile. Although DE demonstrates the best calibration quality in this paper, it also requires the researcher to construct several models that differ in style and structure from the original model. It is also necessary to ensure that these models have favorable classification performance. This may be difficult to achieve in some tasks. Therefore, it is crucial to design an ensemble strategy based only on the original model, as this allows the ensemble strategy to be applied to any model to improve its calibration performance. (2) Secondly, this paper omits a discussion of an uncertainty estimation method called conformal prediction. Compared to traditional uncertainty estimation methods, conformal prediction has rigorous theoretical guarantees, low computational complexity, eliminates the need to assume data distributions, and can be applied as a post-processing method to various trained models. All these require only the data to satisfy the condition of exchangeability. Therefore, we plan to employ conformal prediction methods in fault diagnosis for uncertainty estimation and model calibration tasks. (3) Finally, we will incorporate more variables such as model capacity [37] and dataset size to find out more factors affecting the performance of calibration methods, and explore a universal rule to improve the calibration

quality to meet the requirements of the academic community.

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